6.2.2 Shell Type

The core surrounds the winding is known as shell type transformer. The Figure 6.3 represents the shell type.

The primary and secondary windings are placed on the central limb one above the other. This gives a better magnetic coupling.

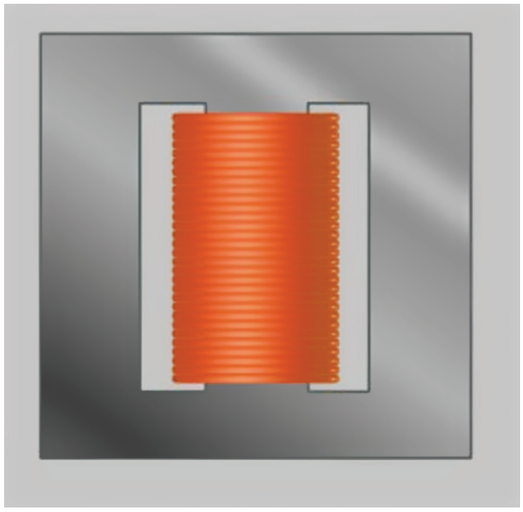


Fig 6.3 *Shell Type Transformer*

Advantages

i) More economical for low voltage.

ii) Low current at the time of no load.

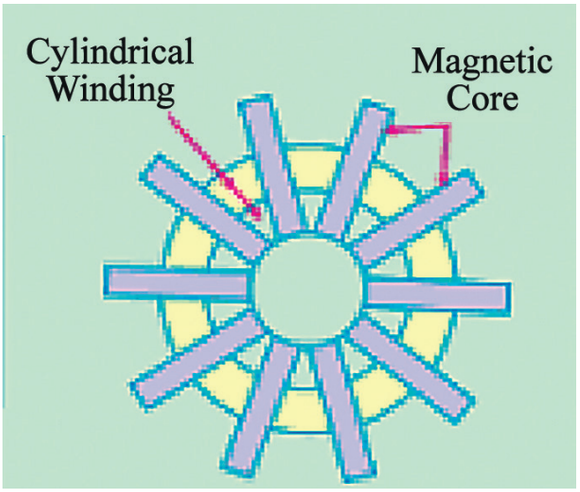
Disadvantages

i) Little complicated to make winding.

ii) Less cooling.

6.2.3 Berry Type

Berry type is similar to shell type. In berry type magnetic path is placed around the coil. Normally this type is not used.



6.2.4 COMPARISON OF CORE AND SHELL TYPE TRANSFORMER

Core Type Transformer	Shell Type Transformer
1. The winding encircles the core.	The core encircles most part of the winding.
2. It has single magnetic circuit.	It has double magnetic circuit.
3. The cylindrical coils are used.	The multilayer disc (or) sandwichs type of coils are used.
4. The coils can be easily removed for maintenance.	The coils cannot be removed easily.

6.3 WORKING PRINCIPLE (OR) OPERATION OF A TRANSFORMER

A Transformer is a static (stationary) apparatus by means of which electric energy in one circuit is transformed

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